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CONVERGENCE OF ACCELERATED GRADIENT METHODS UNDER GENERALIZED SMOOTHNESS AND RELATIVE INEXACTNESS

ABSTRACT

We study first-order methods for convex optimization problems with functions f satisfying the recently proposed ℓ -smoothness condition $\|\nabla^2 f(x)\| \leq \ell(\|\nabla f(x)\|)$, which generalizes the L -smoothness and (L_0, L_1) -smoothness. While accelerated gradient descent (AGD) is known to reach the optimal complexity $\mathcal{O}(\sqrt{LR}/\sqrt{\varepsilon})$ under L -smoothness, where ε is an error tolerance and R is the distance between a starting and an optimal point, existing extensions to ℓ -smoothness either incur extra dependence on the initial gradient, suffer exponential factors in $L_1 R$, or require costly auxiliary sub-routines, leaving open whether an AGD-type $\mathcal{O}(\sqrt{\ell(0)}R/\sqrt{\varepsilon})$ rate is possible for small- ε , even in the (L_0, L_1) -smoothness case.

We resolve this open question. Leveraging a new Lyapunov function and designing new algorithms, we achieve $\mathcal{O}(\sqrt{\ell(0)}R/\sqrt{\varepsilon})$ oracle complexity for small- ε and virtually any ℓ . For instance, for (L_0, L_1) -smoothness, our bound $\mathcal{O}(\sqrt{L_0}R/\sqrt{\varepsilon})$ is provably optimal in the small- ε regime and removes all non-constant multiplicative factors present in prior accelerated algorithms.

1 INTRODUCTION

We focus on optimization problems

$$\min_{x \in \mathbb{R}^d} f(x), \tag{1}$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\infty\}$ is a convex function. We aim to find an ε -solution, $\bar{x} \in \mathbb{R}^d$, such that $f(\bar{x}) - \inf_{x \in \mathbb{R}^d} f(x) \leq \varepsilon$. We define $\mathcal{X} = \{x \in \mathbb{R}^d \mid f(x) < \infty\}$, and assume that $\mathcal{X}$ is an open and d -dimensional convex set, f is smooth on $\mathcal{X}$, and continuous on the closure of $\mathcal{X}$. We define $R := \|x^0 - x^*\|$, where $x^0 \in \mathcal{X}$ is a starting point of numerical methods.

Under the L -smoothness assumption, i.e., $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ or $\|\nabla^2 f(x)\| \leq L$ for all $x, y \in \mathcal{X}$, the problem is well studied. In particular, it is known that one can find an ε -solution after $\mathcal{O}(\sqrt{LR}/\sqrt{\varepsilon})$ gradient calls using the fast/accelerated gradient descent method (AGD) by ?, which is also optimal (??). This result improves the oracle complexity $\mathcal{O}(LR^2/\varepsilon)$ (# of gradient calculations) of gradient descent (GD).

In this work, we investigate the modern ℓ -smoothness assumption (?), which states that $\|\nabla^2 f(x)\| \leq \ell(\|\nabla f(x)\|)$ for all $x \in \mathcal{X}$ (see Assumption 2.1), where ℓ is any non-decreasing, positive, locally Lipschitz function. This generalizes the classical L -smoothness assumption, which corresponds to the special case $\ell(s) = L$. An important example of this framework is the (L_0, L_1) -smoothness condition (?), obtained by setting $\ell(s) = L_0 + L_1 s$, which yields $\|\nabla^2 f(x)\| \leq L_0 + L_1 \|\nabla f(x)\|$ for all $x \in \mathcal{X}$.

Algorithm 1 Accelerated Gradient Descent (AGD) with ℓ -Smoothness

1: **Input:** starting point $x^0 \in \mathcal{X}$, function ℓ from Assumption 2.1, parameters δ and $\bar{R}$
2: Starting from x^0 , run GD from (?) until $f(\bar{x}) - f(x^*) \leq \delta/2$,
 where $\bar{x}$ is the output point of GD
3: Init $y^0 = u^0 = \bar{x}$
4: Set $\Gamma_0 = \delta/\bar{R}^2$
5: Set $\gamma = 1/(2\ell(0))$
6: **for** $k = 0, 1, \dots$ **do**
7: $\alpha_k = \sqrt{\gamma\Gamma_k}$
8: $y^{k+1} = \frac{1}{1+\alpha_k}y^k + \frac{\alpha_k}{1+\alpha_k}u^k - \frac{\gamma}{1+\alpha_k}\nabla f(y^k)$
9: $u^{k+1} = \text{Proj}_{\bar{\mathcal{X}}}\left(u^k - \frac{\alpha_k}{\Gamma_k}\nabla f(y^{k+1})\right)$ ($\bar{\mathcal{X}}$ is the closure of $\mathcal{X}$)
10: $\Gamma_{k+1} = \Gamma_k/(1 + \alpha_k)$
11: **end for**

Assumption 2.1. A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\infty\}$ is ℓ -smooth if f is twice differentiable on $\mathcal{X}$, f is continuous on the closure of $\mathcal{X}$, and there exists a non-decreasing positive locally Lipschitz function $\ell : [0, \infty) \rightarrow (0, \infty)$ such that

$$\|\nabla^2 f(x)\| \leq \ell(\|\nabla f(x)\|) \quad (2)$$

for all $x \in \mathcal{X}$.

The assumption includes L -smoothness when $\ell(s) = L$, (L_0, L_1) -smoothness when $\ell(s) = L_0 + L_1 s$, and (ρ, L_0, L_1) -smoothness, i.e., $\|\nabla^2 f(x)\| \leq L_0 + L_1 \|\nabla f(x)\|^\rho$ for all $x \in \mathcal{X}$, when $\ell(s) = L_0 + L_1 s^\rho$, where $L, L_0, L_1, \rho \geq 0$ are some fixed constants. While Assumption 2.1 requires twice differentiability, the main theorems and algorithms do not directly rely on it. Let us recall the following lemma, which follows from Assumption 2.1:

Lemma 2.2 (?). For all $x, y \in \mathcal{X}$ such that $\|y - x\| \in [0, q_{\max}(\|\nabla f(x)\|)]$, if f is ℓ -smooth (Assumption 2.1), then

$$\|\nabla f(y) - \nabla f(x)\| \leq q^{-1}(\|y - x\|; \|\nabla f(x)\|), \quad (3)$$

where $q(s; a) := \int_0^s \frac{dv}{\ell(a+v)}$, q^{-1} is the inverse of q with respect to s , and $q_{\max}(a) := \int_0^\infty \frac{dv}{\ell(a+v)}$.

Not requiring twice differentiability, we can assume that (3) holds instead of (2). The main reason why we start with (2) is because it is arguably more interpretable. Next, we assume the convexity of f :

Assumption 2.3. A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\infty\}$ is convex and attains the minimum at a (non-unique) $x^* \in \mathbb{R}^d$. We define $R := \|x^0 - x^*\|$, where x^0 is a starting point of numerical methods.

In the theoretical analysis and proofs, it is useful to define the ψ -function:

Definition 2.4 (ψ and ψ^{-1} functions). Let Assumption 2.1 hold. We define the function $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$, and $\psi^{-1} : [0, \psi(\Delta_{\max})) \rightarrow [0, \Delta_{\max})$ as its (standard) inverse, where $\Delta_{\max} \in (0, \infty]$ is the largest constant such that ψ is strictly increasing on¹ $[0, \Delta_{\max})$.

3 SUBQUADRATIC AND QUADRATIC GROWTH OF ℓ

We are ready to present our first result. Consider Algorithm 1, which consists of two phases: first, we run (non-accelerated) GD, and then we run an accelerated version of GD. Later, we will present Algorithm ??, which avoids the first phase. We first state the convergence rate of Algorithm 1 and then discuss and explain it in more detail. We begin by stating a standard result from the theory of accelerated methods (???) concerning auxiliary sequences, which control convergence rates:

¹ $\Delta_{\max} > 0$ due to Lemma B.4.

Theorem 3.1. For any $\Gamma_0 > 0$ and $\gamma \geq 0$, let $\alpha_k \geq \sqrt{\gamma \Gamma_k}$ and $\Gamma_{k+1} = \Gamma_k / (1 + \alpha_k)$ for all $k \geq 0$. Then,

$$\Gamma_{k+1} \leq \frac{9}{\gamma (k+1 - \bar{k})^2} \quad (4)$$

for all $k \geq \bar{k} := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\gamma \Gamma_0}{4} \right), 0 \right\}$.

The following result provides the convergence rate of Algorithm 1 for ℓ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ is strictly increasing, which holds, for instance, under (L_0, L_1) -smoothness.

Theorem 3.2. Suppose that Assumptions 2.1 and 2.3 hold. Let $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ be strictly increasing. Then Algorithm 1 guarantees that

$$f(y^{k+1}) - f(x^*) \leq \Gamma_{k+1} \bar{R}^2 \quad (5)$$

for all $k \geq 0$ with any $\delta \in (0, \infty]$ such that $\ell \left(8\sqrt{\delta \ell(0)} \right) \leq 2\ell(0)$ and any $\bar{R} \geq R := \|x^0 - x^*\|$.

Moreover, for all $k \geq \bar{k} := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\Gamma_0}{8\ell(0)} \right), 0 \right\}$,

$$f(y^{k+1}) - f(x^*) \leq \frac{18\ell(0) \bar{R}^2}{(k+1 - \bar{k})^2}$$

due to Theorem 3.1.

The theorem establishes the desired $1/k^2$ convergence rate of accelerated methods. However, the method enters this regime only after running the GD method and after the initial $\bar{k}$ steps of the accelerated steps. The main and final result, which captures the total oracle complexity, is presented below.

Theorem 3.3. Consider the assumptions and results of Theorem 3.2. The oracle complexity (i.e., the number of gradient calls) required to find an ε -solution is

$$\frac{5\sqrt{\ell(0)}\bar{R}}{\sqrt{\varepsilon}} + k(\delta), \quad (6)$$

for all $\delta \geq 0$ such that $\ell \left(8\sqrt{\delta \ell(0)} \right) \leq 2\ell(0)$, where $k(\delta) := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\delta}{8\ell(0)\bar{R}^2} \right), 0 \right\} + k_{\text{GD}}(\delta)$, $k_{\text{GD}}(\delta)$ is the oracle complexity of GD for finding a point $\bar{x}$ such that $f(\bar{x}) - f(x^*) \leq \delta/2$.

Corollary 3.4. In Theorem 3.3, minimizing over δ and taking $\bar{R} = R := \|x^0 - x^*\|$, the oracle complexity is

$$\frac{5\sqrt{\ell(0)}R}{\sqrt{\varepsilon}} + \underbrace{\min_{\delta \geq 0 : \ell(8\sqrt{\delta \ell(0)}) \leq 2\ell(0)} k(\delta)}_{\text{does not depend on } \varepsilon}. \quad (7)$$

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A EXPERIMENTS

We compare GD (?) and AGD (Algorithm ??) on the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $f(x, y) = e^x + e^{1-x} + \frac{\mu}{2}y^2$, where $\mu = 0.001$. This function is $(3.3 + \mu, 1)$ -smooth and has its minimum at $(0.5, 0)$. Starting at $x^0 = (-6, -5)$, and taking $\bar{R} = 100 \gg R$ and $\Gamma_0 = 100 \gg \frac{2\Delta}{R^2}$ in Algorithm ??, we obtain Figure 1. In this plot, we observe the distinctive accelerated convergence rate of Algorithm ?? with non-monotonic behavior, supporting our theoretical results.

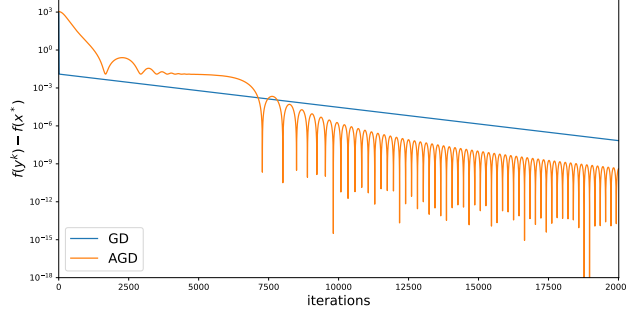


Figure 1: Experiment with $e^x + e^{1-x} + \frac{\mu}{2}y^2$ and $\mu = 0.001$

B AUXILIARY LEMMAS

In the proofs, we use the following useful lemma from (?), which generalizes the key inequality from Theorem 2.1.5 of (?).

Lemma B.1 (?). *For all $x, y \in \mathcal{X}$, if f is ℓ -smooth (Assumption 2.1) and convex (Assumption 2.3), then*

$$\|\nabla f(x) - \nabla f(y)\|^2 \int_0^1 \frac{1-v}{\ell(\|\nabla f(x)\| + \|\nabla f(y)\|v)} dv \leq f(x) - f(y) - \langle \nabla f(y), x - y \rangle. \quad (8)$$

The following lemma ensures that it is “safe” to take steps with proper step sizes.

Lemma B.2 (?). *Under Assumption 2.1, for a fixed $x \in \mathcal{X}$, the point $y = x + th \in \mathcal{X}$ for all $t \in \left[0, \int_0^\infty \frac{dv}{\ell(\|\nabla f(x)\| + v)}\right)$ and $h \in \mathbb{R}^d$ such that $\|h\| = 1$.*

We now prove two important lemmas that allow us to bound the norm $\|\nabla f(y)\|$ given an upper bound on $f(y) - f(x^*)$.

Lemma B.3. [Strictly Increasing ψ] *Under Assumptions 2.1 and 2.3, let $f(y) - f(x^*) \leq \delta$ for some $y \in \mathcal{X}$, $\delta > 0$ and $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ is strictly increasing, then $\|\nabla f(y)\| \leq \psi^{-1}(\delta)$ if $\delta \in \text{im}(\psi)$.*

Proof. Using Lemma B.1 and the fact that ℓ is non-decreasing,

$$\begin{aligned} \delta &\geq f(y) - f(x^*) \geq \|\nabla f(y)\|^2 \int_0^1 \frac{1-v}{\ell(\|\nabla f(y)\| + \|\nabla f(y)\|v)} dv \\ &\geq \frac{\|\nabla f(y)\|^2}{2\ell(4\|\nabla f(y)\|)} = \psi(\|\nabla f(y)\|). \end{aligned}$$

It left to invert ψ to get the result. $\square$

Lemma B.4. [Not Necessarily Strictly Increasing ψ] *Under Assumptions 2.1 and 2.3, let $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ is not necessarily strictly increasing.*

1. There exists the largest $\Delta_{\max} \in (0, \infty]$ such that ψ is strictly increasing on $[0, \Delta_{\max})$.
2. For all $\delta \in [0, \psi(\Delta_{\max}))$, there exists the unique $\Delta_{\text{left}}(\delta) \in [0, \Delta_{\max})$ and the smallest² $\Delta_{\text{right}}(\delta) \in [\Delta_{\max}, \infty]$ such that $\psi(\Delta_{\text{left}}(\delta)) = \delta$ and $\psi(\Delta_{\text{right}}(\delta)) = \delta$.
3. For all $\delta \in [0, \psi(\Delta_{\max}))$, if $\Delta_{\text{right}}(\delta) < \infty$ and $\delta > \bar{\delta} \geq 0$, then $\Delta_{\text{right}}(\bar{\delta}) > \Delta_{\text{right}}(\delta)$.
4. If $f(y) - f(x^*) \leq \delta$ for some $y \in \mathcal{X}$ and $\delta \in [0, \psi(\Delta_{\max}))$, then either $\|\nabla f(y)\| \leq \Delta_{\text{left}}(\delta)$ or $\|\nabla f(y)\| \geq \Delta_{\text{right}}(\delta)$.

Proof. 1. Since ℓ is non-decreasing and locally Lipschitz, there exists $\bar{\Delta}_1 > 0$ such that

$$2\ell(4y) - 2\ell(4x) \leq M(y - x)$$

for all $0 \leq x < y \leq \bar{\Delta}_1$ and for some $M \equiv M(\bar{\Delta}_1, \ell) > 0$. Thus,

$$x^2 2\ell(4y) \leq x^2 2\ell(4x) + Mx^2(y - x). \quad (9)$$

Moreover, there exists $\bar{\Delta}_2 > 0$ such that

$$Mx^2 < (y + x)2\ell(4x)$$

for all $0 \leq x < y \leq \bar{\Delta}_2$ since $2\ell(4x) \geq \ell(0) > 0$, the l.h.s $\mathcal{O}(x^2)$, and the r.h.s. $\Omega(x)$. Combining with (9),

$$x^2 2\ell(4y) < x^2 2\ell(4x) + 2\ell(4x)(y + x)(y - x) = y^2 2\ell(4x)$$

and

$$\frac{x^2}{2\ell(4x)} < \frac{y^2}{2\ell(4y)}$$

for all $0 \leq x < y \leq \min\{\bar{\Delta}_1, \bar{\Delta}_2\}$, meaning that ψ is locally strictly increasing on the interval $[0, \Delta_{\max})$ for some largest $\Delta_{\max} \in (0, \infty]$.

2. $\Delta_{\text{left}}(\delta)$ exists since ψ is locally strictly increasing on the interval $[0, \Delta_{\max})$. On the interval $[\Delta_{\max}, \infty)$, either ψ intersects δ for the first time at $\Delta_{\text{right}}(\delta)$ or we can take $\Delta_{\text{right}}(\delta) = \infty$.

3. Since $\Delta_{\text{right}}(\delta)$ is the first time when ψ intersects δ for $x \in [\Delta_{\max}, \infty)$ and $\delta < \psi(\Delta_{\max})$, then $\psi(x) > \delta$ for all $x \in [\Delta_{\max}, \Delta_{\text{right}}(\delta))$. Thus, if we decrease δ and take $\bar{\delta} < \delta$, then $\Delta_{\text{right}}(\bar{\delta})$ can only increase or stay the same. However, if $\Delta_{\text{right}}(\bar{\delta})$ stays the same, i.e., $\Delta_{\text{right}}(\bar{\delta}) = \Delta_{\text{right}}(\delta)$, then $\Delta_{\text{right}}(\bar{\delta})$ is the first time when ψ intersects $\bar{\delta}$, which is impossible due to the continuity of ψ and the fact that $\Delta_{\text{right}}(\bar{\delta})$ is the first time when ψ intersects $\bar{\delta} < \delta$.

4. Using the same reasoning as in the proof of Lemma B.3:

$$\delta \geq \psi(\|\nabla f(y)\|). \quad (10)$$

Due to the previous properties, either $\|\nabla f(y)\| \leq \Delta_{\text{left}}(\delta)$ or $\|\nabla f(y)\| \geq \Delta_{\text{right}}(\delta)$ because $\psi(x) > \delta$ for all $x \in (\Delta_{\text{left}}(\delta), \Delta_{\text{right}}(\delta))$. $\square$

C RATE OF THE AUXILIARY SEQUENCE

Theorem 3.1. For any $\Gamma_0 > 0$ and $\gamma \geq 0$, let $\alpha_k \geq \sqrt{\gamma\Gamma_k}$ and $\Gamma_{k+1} = \Gamma_k/(1 + \alpha_k)$ for all $k \geq 0$. Then,

$$\Gamma_{k+1} \leq \frac{9}{\gamma(k+1-\bar{k})^2} \quad (4)$$

for all $k \geq \bar{k} := \max\left\{1 + \frac{1}{2} \log_{3/2}\left(\frac{\gamma\Gamma_0}{4}\right), 0\right\}$.

²if the set $\{x \in [\Delta_{\max}, \infty) : \psi(x) = \delta\}$ is empty, then $\Delta_{\text{right}}(\delta) = \infty$

Proof. By the definition of Γ_{k+1} and α_k ,

$$\Gamma_{k+1} \leq \frac{\Gamma_k}{1 + \sqrt{\gamma}\Gamma_k}$$

for all $k \geq 0$. Instead of Γ_k , consider the sequence $\bar{\Gamma}_k$ such that

$$\bar{\Gamma}_{k+1} = \frac{\bar{\Gamma}_k}{1 + \sqrt{\gamma}\bar{\Gamma}_k}$$

for all $k \geq 0$ and $\bar{\Gamma}_0 = \Gamma_0$. Using mathematical induction, notice that $\bar{\Gamma}_{k+1} \geq \Gamma_{k+1}$. Indeed, the function $\frac{x}{1+\sqrt{\gamma}x}$ is increasing³ for all $x \geq 0$ and

$$\Gamma_{k+1} \leq \frac{\Gamma_k}{1 + \sqrt{\gamma}\Gamma_k} \leq \frac{\bar{\Gamma}_k}{1 + \sqrt{\gamma}\bar{\Gamma}_k} = \bar{\Gamma}_{k+1}$$

if $\Gamma_k \leq \bar{\Gamma}_k$. If we bound $\bar{\Gamma}_{k+1}$, then we can bound Γ_{k+1} . Next,

$$\frac{1}{\bar{\Gamma}_{k+1}} - \frac{1}{\bar{\Gamma}_k} = \sqrt{\frac{\gamma}{\bar{\Gamma}_k}}$$

Let us define $t_k := \frac{1}{\bar{\Gamma}_k}$ for all $k \geq 0$, then

$$t_{k+1} - t_k = \sqrt{\gamma t_k}. \quad (11)$$

and

$$(t_{k+1}^{1/2} + t_k^{1/2})(t_{k+1}^{1/2} - t_k^{1/2}) = \sqrt{\gamma t_k} \quad (12)$$

for all $k \geq 0$. We now fix any $k \geq 0$. There are two options:

Option 1: $t_k^{1/2} \leq \frac{\sqrt{\gamma}}{2}$.

In this case, using (11),

$$t_{k+1} = t_k + \sqrt{\gamma t_k} \leq \frac{\gamma}{4} + \frac{\gamma}{2} = \frac{3\gamma}{4}$$

and

$$2\sqrt{\gamma}(t_{k+1}^{1/2} - t_k^{1/2}) \geq \sqrt{\gamma t_k}$$

due to (12). Rearranging the terms,

$$t_{k+1}^{1/2} \geq \frac{3}{2}t_k^{1/2} \geq \left(\frac{3}{2}\right)^{k+1} t_0^{1/2}, \quad (13)$$

where we unroll the recursion since $t_0^{1/2} \leq \dots \leq t_k^{1/2} \leq \frac{\sqrt{\gamma}}{2}$.

Option 2: $t_k^{1/2} > \frac{\sqrt{\gamma}}{2}$.

Using (11),

$$t_{k+1} = t_k + \sqrt{\gamma t_k} \leq t_k + 2t_k \leq 3t_k$$

and

$$3t_k^{1/2}(t_{k+1}^{1/2} - t_k^{1/2}) \geq \sqrt{\gamma t_k}$$

due to (12), which yields

$$t_{k+1}^{1/2} \geq t_k^{1/2} + \frac{\sqrt{\gamma}}{3}. \quad (14)$$

Let $k^* \geq 0$ be the smallest index such that $t_{k^*}^{1/2} > \frac{\sqrt{\gamma}}{2}$. Unrolling (14),

$$t_{k+1}^{1/2} \geq t_{k^*}^{1/2} + (k+1-k^*)\frac{\sqrt{\gamma}}{3} \quad (15)$$

³ $\left(\frac{x}{1+\sqrt{\gamma}x}\right)' = \frac{1+\frac{\sqrt{\gamma}x}{2}}{(1+\sqrt{\gamma}x)^2} > 0$ for all $x \geq 0$.

Algorithm 2 Accelerated Gradient Descent (AGD) with ℓ -Smoothness and Inexact Gradient $g(x) = \nabla f(x) + e(x)$

1: **Input:** starting point $x^0 \in \mathcal{X}$, function ℓ from Assumption 2.1, parameters δ and $\bar{R}$
2: Starting from x^0 , run GD from (?) until $f(\bar{x}) - f(x^*) \leq \delta/2$,
where $\bar{x}$ is the output point of GD
3: Init $y^0 = u^0 = \bar{x}$
4: Set $\Gamma_0 = \delta/\bar{R}^2$
5: Set $\gamma = 1/(2\ell(0))$
6: **for** $k = 0, 1, \dots$ **do**
7: $\alpha_k = \sqrt{\gamma\Gamma_k}$
8: $y^{k+1} = \frac{1}{1+\alpha_k}y^k + \frac{\alpha_k}{1+\alpha_k}u^k - \frac{\gamma}{1+\alpha_k}g(y^k)$
9: $u^{k+1} = \text{Proj}_{\bar{\mathcal{X}}}\left(u^k - \frac{\alpha_k}{\Gamma_k}g(y^{k+1})\right)$ ($\bar{\mathcal{X}}$ is the closure of $\mathcal{X}$)
10: $\Gamma_{k+1} = \Gamma_k/(1 + \alpha_k)$
11: **end for**

for all $k \geq k^*$. If $k^* = 0$, then

$$t_{k+1}^{1/2} \geq (k+1) \frac{\sqrt{\gamma}}{3}. \quad (16)$$

Otherwise, by the definition of k^* ,

$$\left(\frac{3}{2}\right)^{k^*-1} t_0^{1/2} \stackrel{(13)}{\leq} t_{k^*-1}^{1/2} \leq \frac{\sqrt{\gamma}}{2},$$

which yields

$$k^* \leq 1 + \frac{1}{2} \log_{3/2} \left(\frac{\gamma}{4t_0} \right)$$

and

$$t_{k+1}^{1/2} \geq \left(k+1 - \left(1 + \frac{1}{2} \log_{3/2} \left(\frac{\gamma}{4t_0} \right) \right) \right) \frac{\sqrt{\gamma}}{3}, \quad (17)$$

due to (15). Combining the cases with $k^* = 0$ and $k^* > 0$, we get

$$t_{k+1}^{1/2} \geq (k+1 - \bar{k}) \frac{\sqrt{\gamma}}{3} \quad (18)$$

for all $k \geq \bar{k} := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\gamma}{4t_0} \right), 0 \right\}$. It left to recall that $t_k = 1/\bar{\Gamma}_k$ and $\bar{\Gamma}_k \geq \Gamma_k$ for all $k \geq 0$ to obtain (4). □

D MAIN DESCENT LEMMA

Lemma D.1. Suppose that Assumptions 2.1 and 2.3 hold. Consider Algorithm 1 up to the k^{th} iteration and the following virtual steps:

$$\begin{aligned} \alpha_k(\gamma) &\equiv \alpha_{k,\gamma} = \sqrt{\gamma\Gamma_k}, \\ y^{k+1}(\gamma) &\equiv y_{\gamma}^{k+1} = \frac{1}{1+\alpha_{k,\gamma}}y^k + \frac{\alpha_{k,\gamma}}{1+\alpha_{k,\gamma}}u^k - \frac{\gamma}{1+\alpha_{k,\gamma}}g(y^k), \\ u^{k+1}(\gamma) &\equiv u_{\gamma}^{k+1} = \text{Proj}_{\bar{\mathcal{X}}}\left(u^k - \frac{\alpha_{k,\gamma}}{\Gamma_k}g(y_{\gamma}^{k+1})\right), \\ \Gamma_{k+1}(\gamma) &\equiv \Gamma_{k+1,\gamma} = \Gamma_k/(1 + \alpha_{k,\gamma}), \end{aligned} \quad (19)$$

where $0 \leq \gamma \leq \frac{1}{\ell(2\|\nabla f(y^k)\|)}$ is a free parameter, $y^k \in \mathcal{X}$, and $u^k \in \bar{\mathcal{X}}$. Then, the steps (19) are well-defined, $y_{\gamma}^{k+1} \in \mathcal{X}$, and $u_{\gamma}^{k+1} \in \bar{\mathcal{X}}$, and

$$(1 + \alpha_{k,\gamma})(f(y_{\gamma}^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_{\gamma}^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right)$$

$$\leq \frac{1}{2} \left(\gamma - \frac{1}{\ell(2 \|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} \right) \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2.$$

Proof. Consider the difference

$$f(y_\gamma^{k+1}) - f(x^*) + \frac{\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right). \quad (20)$$

Rearranging the terms, we get

$$\begin{aligned} & f(y_\gamma^{k+1}) - f(x^*) + \frac{\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\ &= -(f(y^k) - f(y_\gamma^{k+1})) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \\ &\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k \rangle \\ &\quad + \frac{\Gamma_{k+1,\gamma} - \Gamma_k}{2} \|u_\gamma^{k+1} - x^*\|^2 + \frac{\Gamma_k}{2} (\|u_\gamma^{k+1} - x^*\|^2 - \|u^k - x^*\|^2). \end{aligned}$$

Since $\Gamma_k = (1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}$,

$$\begin{aligned} & f(y_\gamma^{k+1}) - f(x^*) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\ &= -(f(y^k) - f(y_\gamma^{k+1})) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \\ &\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k \rangle \\ &\quad + \frac{\Gamma_k}{2} (\|u_\gamma^{k+1} - x^*\|^2 - \|u^k - x^*\|^2). \end{aligned}$$

Due to $\|a\|^2 - \|a + b\|^2 = -\|b\|^2 - 2\langle a, b \rangle$ for all $a, b \in \mathbb{R}^d$,

$$\begin{aligned} & f(y_\gamma^{k+1}) - f(x^*) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\ &= -(f(y^k) - f(y_\gamma^{k+1})) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \\ &\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k \rangle \\ &\quad + \frac{\Gamma_k}{2} \left(-\|u^k - u_\gamma^{k+1}\|^2 - 2\langle u_\gamma^{k+1} - x^*, u^k - u_\gamma^{k+1} \rangle \right). \end{aligned} \quad (21)$$

Consider the last inner product:

$$\begin{aligned} & -\langle u_\gamma^{k+1} - x^*, u^k - u_\gamma^{k+1} \rangle \\ &= \left\langle u_\gamma^{k+1} - x^*, \left(u^k - \frac{\alpha_{k,\gamma}}{\Gamma_k} g(y_\gamma^{k+1}) \right) - u^k \right\rangle + \left\langle u_\gamma^{k+1} - x^*, u_\gamma^{k+1} - \left(u^k - \frac{\alpha_{k,\gamma}}{\Gamma_k} g(y_\gamma^{k+1}) \right) \right\rangle. \end{aligned}$$

Using $u_\gamma^{k+1} = \text{Proj}_{\tilde{\mathcal{X}}} \left(u^k - \frac{\alpha_{k,\gamma}}{\Gamma_k} g(y_\gamma^{k+1}) \right)$ and the projection property $\langle \text{Proj}_{\tilde{\mathcal{X}}}(y) - x, \text{Proj}_{\tilde{\mathcal{X}}}(y) - y \rangle \leq 0$ for all $y \in \mathbb{R}^d, x \in \tilde{\mathcal{X}}$, we have

$$-\langle u_\gamma^{k+1} - x^*, u^k - u_\gamma^{k+1} \rangle \leq \left\langle u_\gamma^{k+1} - x^*, \left(u^k - \frac{\alpha_{k,\gamma}}{\Gamma_k} g(y_\gamma^{k+1}) \right) - u^k \right\rangle = -\left\langle u_\gamma^{k+1} - x^*, \frac{\alpha_{k,\gamma}}{\Gamma_k} \nabla f(y_\gamma^{k+1}) \right\rangle.$$

Substituting to (21),

$$\begin{aligned} & f(y_\gamma^{k+1}) - f(x^*) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\ &= -(f(y^k) - f(y_\gamma^{k+1})) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \\ &\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k \rangle \\ &\quad + \frac{\Gamma_k}{2} \left(-\|u^k - u_\gamma^{k+1}\|^2 - 2\left\langle u_\gamma^{k+1} - x^*, \frac{\alpha_{k,\gamma}}{\Gamma_k} g(y_\gamma^{k+1}) \right\rangle \right) \end{aligned}$$

$$\begin{aligned}
&= -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k \rangle \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 \\
&\quad - \alpha_{k,\gamma} \langle u_\gamma^{k+1} - x^*, g(y_\gamma^{k+1}) \rangle \\
&= -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k - \alpha_{k,\gamma}(u_\gamma^{k+1} - y_\gamma^{k+1}) \rangle \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 \\
&\quad - \alpha_{k,\gamma} \langle y_\gamma^{k+1} - x^*, \nabla f(y_\gamma^{k+1}) \rangle + \alpha_{k,\gamma} \langle x^* - u_\gamma^{k+1}, e_\gamma^{k+1} \rangle.
\end{aligned}$$

In the last two equalities, we rearranged terms. Using the convexity of f , we have $-(f(y_\gamma^{k+1}) - f(x^*)) \geq -\langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - x^* \rangle$ and

$$\begin{aligned}
&(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\
&\leq -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \langle \nabla f(y_\gamma^{k+1}), y_\gamma^{k+1} - y^k - \alpha_{k,\gamma}(u_\gamma^{k+1} - y_\gamma^{k+1}) \rangle \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 + \alpha_{k,\gamma} \langle x^* - u_\gamma^{k+1}, e_\gamma^{k+1} \rangle \\
&= -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \langle \nabla f(y_\gamma^{k+1}), (1 + \alpha_{k,\gamma})y_\gamma^{k+1} - y^k - \alpha_{k,\gamma}u_\gamma^{k+1} \rangle \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 + \alpha_{k,\gamma} \langle x^* - u_\gamma^{k+1}, e_\gamma^{k+1} \rangle.
\end{aligned}$$

In the last equality, we rearranged terms. Recall that

$$(1 + \alpha_{k,\gamma})y_\gamma^{k+1} - y^k = \alpha_{k,\gamma}u^k - \gamma g(y^k).$$

Thus,

$$\begin{aligned}
&(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\
&\leq -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \alpha_{k,\gamma} \langle \nabla f(y_\gamma^{k+1}), u^k - u_\gamma^{k+1} \rangle - \gamma \langle \nabla f(y_\gamma^{k+1}), \nabla f(y^k) \rangle \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 \\
&= -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \alpha_{k,\gamma} \langle \nabla f(y_\gamma^{k+1}), u^k - u_\gamma^{k+1} \rangle \\
&\quad - \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1})\|^2 - \frac{\gamma}{2} \|\nabla f(y^k)\|^2 + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2 \\
&\quad - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2,
\end{aligned}$$

where we use $-\langle a, b \rangle = \frac{1}{2} \|a - b\|^2 - \frac{1}{2} \|a\|^2 - \frac{1}{2} \|b\|^2$ for all $a, b \in \mathbb{R}^d$. Using Young's inequality,

$$\begin{aligned}
&(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\
&\leq -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
&\quad + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1})\|^2 + \frac{\alpha_{k,\gamma}^2}{2\gamma} \|u^k - u_\gamma^{k+1}\|^2
\end{aligned}$$

$$\begin{aligned}
& -\frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1})\|^2 - \frac{\gamma}{2} \|\nabla f(y^k)\|^2 + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2 \\
& - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 \\
& = -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
& + \frac{\alpha_{k,\gamma}^2}{2\gamma} \|u^k - u_\gamma^{k+1}\|^2 - \frac{\Gamma_k}{2} \|u^k - u_\gamma^{k+1}\|^2 \\
& - \frac{\gamma}{2} \|\nabla f(y^k)\|^2 + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2,
\end{aligned}$$

where the terms $\frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1})\|^2$ are cancelled out. Since $\alpha_{k,\gamma} = \sqrt{\gamma\Gamma_k}$, the terms with $\|u^k - u_\gamma^{k+1}\|^2$ are also cancelled out and

$$\begin{aligned}
& (1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\
& \leq -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) \\
& - \frac{\gamma}{2} \|\nabla f(y^k)\|^2 + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2 \\
& \leq -(f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle) + \frac{\gamma}{2} \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2, \tag{22}
\end{aligned}$$

where the last inequality due to $\frac{\gamma}{2} \|\nabla f(y^k)\|^2 \geq 0$. Using Lemma B.1, we get

$$\begin{aligned}
& f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \\
& \geq \|\nabla f(y^k) - \nabla f(y_\gamma^{k+1})\|^2 \int_0^1 \frac{1-v}{\ell(\|\nabla f(y^k)\| + \|\nabla f(y^k) - \nabla f(y_\gamma^{k+1})\|v)} dv \\
& \geq \|\nabla f(y^k) - \nabla f(y_\gamma^{k+1})\|^2 \frac{1}{2\ell(\|\nabla f(y^k)\| + \|\nabla f(y^k) - \nabla f(y_\gamma^{k+1})\|)},
\end{aligned}$$

where we use that ℓ is non-decreasing and bounded the term in the denominator by the maximum possible value with $v = 1$. Using triangle's inequality,

$$f(y^k) - f(y_\gamma^{k+1}) - \langle \nabla f(y_\gamma^{k+1}), y^k - y_\gamma^{k+1} \rangle \geq \|\nabla f(y^k) - \nabla f(y_\gamma^{k+1})\|^2 \frac{1}{2\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)},$$

Substituting to (22),

$$\begin{aligned}
& (1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\
& \leq \frac{1}{2} \left(\gamma - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} \right) \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2.
\end{aligned}$$

□

E CONVERGENCE THEOREMS

E.1 SUBQUADRATIC AND QUADRATIC GROWTH OF ℓ

Lemma E.1. Under Assumptions 2.1 and 2.3, let $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ is strictly increasing, $f(y) - f(x^*) \leq \delta$ for some $y \in \mathcal{X}$, and any $\delta \in (0, \infty]$ such that $\ell\left(8\sqrt{\delta\ell(0)}\right) \leq 2\ell(0)$, then $\ell(4\|\nabla f(y)\|) \leq 2\ell(0)$.

Proof. With this choice of δ , we get

$$\ell(4\|\nabla f(y)\|) \leq 2\ell(0)$$

because, due to $f(y) - f(x^*) \leq \delta$ and Lemma B.3,

$$\ell(4\|\nabla f(y^0)\|) \leq \ell(4\psi^{-1}(\delta))$$

and

$$\begin{aligned} \ell(4\psi^{-1}(\delta)) \leq 2\ell(0) &\Leftrightarrow \frac{(\psi^{-1}(\delta))^2}{4\ell(0)} \leq \frac{(\psi^{-1}(\delta))^2}{2\ell(4\psi^{-1}(\delta))} \stackrel{\psi(\psi^{-1}(\delta))=\delta}{=} \frac{(\psi^{-1}(\delta))^2}{4\ell(0)} \leq \delta \Leftrightarrow \psi^{-1}(\delta) \leq 2\sqrt{\delta\ell(0)} \\ &\Leftrightarrow \delta \leq \psi(2\sqrt{\delta\ell(0)}) \Leftrightarrow \delta \leq \frac{2\delta\ell(0)}{\ell(8\sqrt{\delta\ell(0)})} \Leftrightarrow \ell(8\sqrt{\delta\ell(0)}) \leq 2\ell(0). \end{aligned} \quad (23)$$

□

Theorem 3.2. Suppose that Assumptions 2.1 and 2.3 hold. Let $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\psi(x) = \frac{x^2}{2\ell(4x)}$ be strictly increasing. Then Algorithm 1 guarantees that

$$f(y^{k+1}) - f(x^*) \leq \Gamma_{k+1} \bar{R}^2 \quad (5)$$

for all $k \geq 0$ with any $\delta \in (0, \infty]$ such that $\ell(8\sqrt{\delta\ell(0)}) \leq 2\ell(0)$ and any $\bar{R} \geq R := \|x^0 - x^*\|$.

Moreover, for all $k \geq \bar{k} := \max\left\{1 + \frac{1}{2} \log_{3/2}\left(\frac{\Gamma_0}{8\ell(0)}\right), 0\right\}$,

$$f(y^{k+1}) - f(x^*) \leq \frac{18\ell(0) \bar{R}^2}{(k+1-\bar{k})^2}$$

due to Theorem 3.1.

Proof. In our proof, we define the Lyapunov function $V_k := f(y^k) - f(x^*) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2$. After running GD, we get $\ell(4\|\nabla f(y^0)\|) \leq 2\ell(0)$ due to Lemma E.1 and the choice of δ . Trivially, $V_0 \leq V_0$. Due to $f(y^0) - f(x^*) \leq \frac{\delta}{2}$ in Alg. 1 and $\|y^0 - x^*\| \leq \|x^0 - x^*\|$ (GD is monotonic; (?) [Lemma I.2]),

$$\begin{aligned} V_0 &= f(y^0) - f(x^*) + \frac{\Gamma_0}{2} \|y^0 - x^*\|^2 \leq \frac{\delta}{2} + \frac{\Gamma_0}{2} \|y^0 - x^*\|^2 \\ &\leq \frac{\delta}{2} + \frac{\Gamma_0}{2} \|x^0 - x^*\|^2 \leq \delta \end{aligned} \quad (24)$$

since $\Gamma_0 = \frac{\delta}{\bar{R}^2}$ and $\bar{R} \geq \|x^0 - x^*\|$. Using mathematical induction, we assume that $\ell(4\|\nabla f(y^k)\|) \leq 2\ell(0)$ and $V_k \leq \left(\prod_{i=0}^{k-1} \frac{1}{1+\alpha_i}\right) V_0$ for some $k \geq 0$.

Consider Lemma D.1 and the steps (19). Then,

$$\begin{aligned} &(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2\right) \\ &\leq \frac{1}{2} \left(\gamma - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} \right) \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2, \end{aligned}$$

where $0 \leq \gamma \leq \frac{1}{\ell(2\|\nabla f(y^k)\|)}$ is a free parameter. Let us take the smallest γ such that

$$g(\gamma) := \gamma - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} = 0$$

and denote it as γ^* . Such a choice exists because $g(\gamma)$ is continuous for all $\gamma \geq 0$ as a composition of continuous functions (y_γ^{k+1} is a continuous function of γ), $g(0) = -\frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_0^{k+1})\|)} < 0$, and

$$g(\bar{\gamma}) = \bar{\gamma} - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_{\bar{\gamma}}^{k+1})\|)} \geq \bar{\gamma} - \frac{1}{\ell(2\|\nabla f(y^k)\|)} = 0$$

for $\bar{\gamma} = \frac{1}{\ell(2\|\nabla f(y^k)\|)}$. Note that $\gamma^* \leq \frac{1}{\ell(2\|\nabla f(y^k)\|)}$. For all $\gamma \leq \gamma^*$, $g(\gamma) \leq 0$ and

$$\begin{aligned} & (1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 \\ & \leq (f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 =: V_k, \end{aligned} \quad (25)$$

which ensures that

$$f(y_\gamma^{k+1}) - f(x^*) \leq V_k.$$

Recall that $V_k \leq V_0 \stackrel{(24)}{\leq} \delta$. It means that

$$f(y_\gamma^{k+1}) - f(x^*) \leq \delta$$

and

$$\ell(4\|\nabla f(y_\gamma^{k+1})\|) \leq 2\ell(0)$$

for all $\gamma \leq \gamma^*$ due to Lemma E.1. Therefore, by the definition of γ^* and using $\ell(4\|\nabla f(y^k)\|) \leq 2\ell(0)$,

$$\gamma^* = \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_{\gamma^*}^{k+1})\|)} \geq \frac{1}{\max\{\ell(4\|\nabla f(y^k)\|), \ell(4\|\nabla f(y_{\gamma^*}^{k+1})\|)\}} \geq \frac{1}{2\ell(0)},$$

meaning that we can take $\gamma = \frac{1}{2\ell(0)}$ and (25) holds:

$$(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 \leq V_k.$$

Notice that $\alpha_{k,\gamma} = \alpha_k$, $y_\gamma^{k+1} = y^{k+1}$, $\Gamma_{k+1,\gamma} = \Gamma_{k+1}$, and $u_\gamma^{k+1} = u^{k+1}$ with $\gamma = \frac{1}{2\ell(0)}$. Therefore,

$$(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 = (1 + \alpha_k)V_{k+1},$$

$$\ell(4\|\nabla f(y^{k+1})\|) \leq 2\ell(0),$$

and

$$V_{k+1} \leq \frac{1}{1 + \alpha_k} V_k \leq \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right) V_0,$$

We have proved the next step of the induction. Finally, for all $k \geq 0$,

$$\begin{aligned} f(y^{k+1}) - f(x^*) & \leq V_{k+1} \leq \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right) \left(f(y^0) - f(x^*) + \frac{\Gamma_0}{2} \|y^0 - x^*\|^2 \right) \\ & = \Gamma_0 \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right) \left(\frac{1}{\Gamma_0} (f(y^0) - f(x^*)) + \frac{1}{2} \|y^0 - x^*\|^2 \right). \end{aligned}$$

Since $f(y^0) - f(x^*) \leq \frac{\delta}{2}$, $\|y^0 - x^*\|^2 \leq \|x^0 - x^*\|^2 \leq \bar{R}^2$, and $\Gamma_{k+1} = \Gamma_0 \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right)$,

$$f(y^{k+1}) - f(x^*) \leq \Gamma_{k+1} \left(\frac{\delta}{2\Gamma_0} + \frac{1}{2} \bar{R}^2 \right) = \Gamma_{k+1} \bar{R}^2$$

because $\Gamma_0 = \frac{\delta}{\bar{R}^2}$. □

Theorem 3.3. Consider the assumptions and results of Theorem 3.2. The oracle complexity (i.e., the number of gradient calls) required to find an ε -solution is

$$\frac{5\sqrt{\ell(0)\bar{R}}}{\sqrt{\varepsilon}} + k(\delta), \quad (6)$$

for all $\delta \geq 0$ such that $\ell(8\sqrt{\delta\ell(0)}) \leq 2\ell(0)$, where $k(\delta) := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\delta}{8\ell(0)\bar{R}^2} \right), 0 \right\} + k_{\text{GD}}(\delta)$, $k_{\text{GD}}(\delta)$ is the oracle complexity of GD for finding a point $\bar{x}$ such that $f(\bar{x}) - f(x^*) \leq \delta/2$.

Proof. At the beginning, we run GD, which takes $k_{\text{GD}}(\delta)$ iterations (i.e., gradient evaluations). Next, using Theorem 3.1 and the choice of $\gamma = \frac{1}{2\ell(0)}$,

$$\Gamma_{k+1} \leq \frac{18\ell(0)}{(k+1-\bar{k})^2}$$

for all $k \geq \bar{k} := \max \left\{ 1 + \frac{1}{2} \log_{3/2} \left(\frac{\Gamma_0}{8\ell(0)} \right), 0 \right\}$. Taking

$$k \geq \frac{5\sqrt{\ell(0)\bar{R}}}{\sqrt{\varepsilon}} + \bar{k},$$

we get $f(y^{k+1}) - f(x^*) \leq \varepsilon$ due to Theorem 3.2. $\square$

E.2

*

Proof. In our proof, we define the Lyapunov function $V_k := f(y^k) - f(x^*) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2$. Trivially, $V_0 \leq V_0$ and

$$V_0 = f(y^0) - f(x^*) + \frac{\Gamma_0}{2} \|y^0 - x^*\|^2 \leq \Gamma_0 R^2 \leq \Gamma_0 \bar{R}^2 \quad (26)$$

when $\Gamma_0 \geq \frac{2(f(y^0) - f(x^*))}{\|y^0 - x^*\|^2} = \frac{2(f(y^0) - f(x^*))}{\|y^0 - x^*\|^2}$ and $\bar{R} \geq R$. Moreover,

$$f(y^0) - f(x^*) \leq \Gamma_0 \bar{R}^2.$$

Due to Lemma B.3,

$$\|\nabla f(y^0)\| \leq \psi^{-1}(\Gamma_0 \bar{R}^2).$$

Using mathematical induction, we assume that $\|\nabla f(y^k)\| \leq \psi^{-1}(\Gamma_k \bar{R}^2)$ and $V_k \leq \left(\prod_{i=0}^{k-1} \frac{1}{1+\alpha_i} \right) V_0$ for some $k \geq 0$.

Consider Lemma D.1 and the steps (19). Then,

$$\begin{aligned} & (1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 - \left((f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 \right) \\ & \leq \frac{1}{2} \left(\gamma - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} \right) \|\nabla f(y_\gamma^{k+1}) - \nabla f(y^k)\|^2, \end{aligned}$$

where $0 \leq \gamma \leq \frac{1}{\ell(2\|\nabla f(y^k)\|)}$ is a free parameter. Let us take the smallest γ such that

$$g(\gamma) := \gamma - \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_\gamma^{k+1})\|)} = 0$$

and denote it as γ^* (exists similarly to the proof of Theorem 3.2 and $\gamma^* \leq \frac{1}{\ell(2\|\nabla f(y^k)\|)}$). For all $\gamma \leq \gamma^*$, $g(\gamma) \leq 0$ and

$$\begin{aligned} & (1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 \\ & \leq (f(y^k) - f(x^*)) + \frac{\Gamma_k}{2} \|u^k - x^*\|^2 =: V_k. \end{aligned} \quad (27)$$

Recall that

$$V_k \leq \left(\prod_{i=0}^{k-1} \frac{1}{1+\alpha_i} \right) V_0 = \frac{\Gamma_k}{\Gamma_0} V_0 \stackrel{(26)}{\leq} \Gamma_k \bar{R}^2.$$

Therefore,

$$f(y_\gamma^{k+1}) - f(x^*) \stackrel{(27)}{\leq} \frac{\Gamma_k \bar{R}^2}{1 + \alpha_{k,\gamma}}$$

and

$$\|\nabla f(y_\gamma^{k+1})\| \leq \psi^{-1} \left(\frac{\Gamma_k \bar{R}^2}{1 + \alpha_{k,\gamma}} \right) \leq \psi^{-1} (\Gamma_k \bar{R}^2) \quad (28)$$

for all $\gamma \leq \gamma^*$ due to Lemma B.3. Therefore, by the definition of γ^* and using $\|\nabla f(y^k)\| \leq \psi^{-1}(\Gamma_k \bar{R}^2)$,

$$\gamma^* = \frac{1}{\ell(2\|\nabla f(y^k)\| + \|\nabla f(y_{\gamma^*}^{k+1})\|)} \geq \frac{1}{\max\{\ell(4\|\nabla f(y^k)\|), \ell(4\|\nabla f(y_{\gamma^*}^{k+1})\|)\}} \geq \frac{1}{\ell(4\psi^{-1}(\Gamma_k \bar{R}^2))},$$

meaning that we can take $\gamma_k = \frac{1}{\ell(4\psi^{-1}(\Gamma_k \bar{R}^2))}$ and (25) holds:

$$(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 \leq V_k.$$

Notice that $\alpha_{k,\gamma} = \alpha_k$, $y_\gamma^{k+1} = y^{k+1}$, $\Gamma_{k+1,\gamma} = \Gamma_{k+1}$, and $u_\gamma^{k+1} = u^{k+1}$ with $\gamma = \frac{1}{\ell(4\psi^{-1}(\Gamma_k \bar{R}^2))}$.

Therefore, $(1 + \alpha_{k,\gamma})(f(y_\gamma^{k+1}) - f(x^*)) + \frac{(1 + \alpha_{k,\gamma})\Gamma_{k+1,\gamma}}{2} \|u_\gamma^{k+1} - x^*\|^2 = (1 + \alpha_k)V_{k+1}$,

$$\|\nabla f(y^{k+1})\| \stackrel{(28)}{\leq} \psi^{-1} \left(\frac{\Gamma_k \bar{R}^2}{1 + \alpha_k} \right) = \psi^{-1} (\Gamma_{k+1} \bar{R}^2)$$

and

$$V_{k+1} \leq \frac{1}{1 + \alpha_k} V_k \leq \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right) V_0,$$

We have proved the next step of the induction. Finally, for all $k \geq 0$,

$$f(y^{k+1}) - f(x^*) \leq V_{k+1} \leq \Gamma_{k+1} \left(\frac{1}{\Gamma_0} (f(y^0) - f(x^*)) + \frac{1}{2} \|y^0 - x^*\|^2 \right) \leq \Gamma_{k+1} \|x^0 - x^*\|^2$$

because $\Gamma_0 \geq \frac{2(f(y^0) - f(x^*))}{\|y^0 - x^*\|^2}$, $\Gamma_{k+1} = \Gamma_0 \left(\prod_{i=0}^k \frac{1}{1 + \alpha_i} \right)$, and $y^0 = x^0$. $\square$